RAHUL UPADHYAY

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PROFESSIONAL SUMMARY

AI & Backend Engineer specializing in building scalable Retrieval-Augmented Generation (RAG) pipelines, production-ready REST APIs, and microservices architecture. Proficient in Python, FastAPI, Node.js, Express.js, Vector Databases (Qdrant), Redis, Docker, and AWS. Experienced in building secure authentication systems, high-concurrency backend services, and AI agent workflows.

EDUCATION

G.L. BAJAJ INSTITUTE OF TECHNOLOGY AND MANAGEMENT **Greater Noida, India**
Bachelor of Technology - Computer Science and Engineering with Specialization in AI CGPA: 7.25 | Sept. 2023 - July 2027

MERRY CITY SCHOOL **Varanasi, India**
Intermediate (2020 - 2022) | *High School (2018 - 2020)*

TECHNICAL SKILLS AND INTERESTS

- **AI / Machine Learning & LLMs:** Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), Agentic Workflows, PyTorch, LangChain, Prompt Engineering, Vector Databases (Qdrant, Pinecone), Fine-Tuning
- **Programming Languages:** Python, Java, JavaScript
- **Frameworks & Libraries:** FastAPI, Node.js, Express.js, PyTorch, OpenCV, NumPy, Pandas, Scikit-Learn
- **Databases & Caching:** MongoDB, SQL (PostgreSQL/MySQL), Redis (Caching & Distributed Locking), Vector Search
- **Backend, Cloud & DevOps:** REST API Design, Microservices, Docker, Docker Compose, AWS (EC2, S3), Git, GitHub, Rate Limiting, JWT Authentication, System Design
- **Area of Interest:** Artificial Intelligence, Backend Engineering, Cloud Systems, Competitive Programming

PROJECTS

DevOps Copilot — Infrastructure & Log Analytics RAG Platform ***(Python, FastAPI, Qdrant, Gemini LLM, PostgreSQL, Redis, JWT, Docker)***

- **Vector RAG Search Architecture:** Engineered an end-to-end RAG-powered log analytics system parsing live server error logs (Kubernetes, AWS, Nginx) and executing dense vector similarity search in Qdrant against indexed DevOps documentation with <100ms retrieval latency.
- **AI Root-Cause Synthesis:** Integrated Google Gemini LLM with prompt engineering and context injection to synthesize automated root-cause analysis reports and exact terminal fix commands for system failures.
- **Security & Authentication:** Secured REST API endpoints using JWT Bearer authentication, role-based access control, and Bcrypt password hashing; persisted user scan history and audit reports in PostgreSQL.
- **Containerized Microservices:** Containerized FastAPI backend, Qdrant vector database, and Redis service using Docker & Docker Compose for seamless single-command cloud deployment.

Scalable Movie Ticket Booking Backend System ***(Node.js, Express.js, PostgreSQL, Redis, JWT Authentication, REST API)***

- **High-Concurrency Ticket Locking:** Architected a high-throughput movie ticket booking REST API utilizing Redis distributed locks and atomic transactions to prevent double-booking of seats during peak traffic booking surges.
- **RESTful Services & Auth:** Implemented modular REST endpoints for user authentication (JWT + Bcrypt), movie/showtime management, and ticket reservations with strict request schema validation.
- **Database Design & ACID Compliance:** Designed an optimized PostgreSQL relational schema with indexed queries, foreign key constraints, and ACID-compliant transaction handling for seat status management.
- **Automated Expiration Queue:** Built an asynchronous Redis background worker queue to automatically release reserved seats if payment confirmation is not received within a 5-minute timeout window.

ACHIEVEMENTS

- **Smart India Hackathon 2025:** Advanced to national-level SIH as one of the top 45 teams (selected from 350+ internal teams at the college level) for building innovative AI pattern recognition models.
- **LeetCode:** Solved 250+ coding questions on LeetCode demonstrating strong data structures, algorithms, and computational problem-solving proficiency.